

Exact Real Chamber Spectra of Explicit Keller Maps

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Abstract

Migus classified the generic degrees of real polynomial Keller maps with non-dense Euclidean image: in every dimension at least three they are exactly the even integers at least four. We refine the realization side inside one explicit weighted family. For every $N \geq 3$ we give a rational Keller map $F_N : \mathbb{A}^3 \rightarrow \mathbb{A}^3$ of generic degree N whose complete regular real fibers have exactly

$$N, N-2, N-4, \dots, N \bmod 2$$

points, and no other cardinalities. Every value occurs on a nonempty open target chamber, every chamber contains a rational target, and selected chambers with the displayed cardinalities form a chain whose successive walls are smooth folds of the inverse polynomial pencil. In even degree the last chamber is disjoint from the image, recovering the non-dense examples in Migus's classification with their full real sheet spectrum.

1 Context and statement

A real polynomial map with nonzero constant Jacobian determinant will be called a real Keller map. Migus proved that, for $m \geq 3$, the possible generic degrees of such maps $\mathbb{R}^m \rightarrow \mathbb{R}^m$ with non-dense Euclidean image are precisely the even integers $N \geq 4$ [3]. His lower bound combines conjugation parity with Campbell's exclusion of generic degree two; his construction proves that every permitted degree occurs.

The examples used here belong to the weighted-lift construction developed by Gallagher, who also gave the first explicit quartic member with an empty real fiber chamber and a member of every generic degree $N \geq 3$ [2]. All-degree families also occur in the earlier base manuscript [1]. The point of this note is different: for one explicit rational member in every degree, we determine all real sheet numbers on the natural complete regular locus, give rational witnesses, and exhibit the full parity chain by adjacent fold crossings.

We first define the maps. For $N = 2k + 1$ and $N = 2k$, respectively, put

$$\begin{aligned} S_{2k+1} &= \{2\} \cup \{j, 1-j : 3 \leq j \leq k+1\}, \\ S_{2k} &= \{3, 4\} \cup \{j, 1-j : 5 \leq j \leq k+2\}, \\ G_N(W) &= W(W+1) \prod_{\rho \in S_N} (W - \rho), \end{aligned} \tag{1.1}$$

$$\lambda_N = G'_N(0) - G'_N(1), \quad H_N(W) = \frac{G_N(W) - G'_N(0)W}{\lambda_N}.$$

Empty ranges are omitted, so these formulas include $N = 3, 4$. The roots of G_N are N distinct integers, and

$$\lambda_{2k+1} = (-1)^k k(k!)^2, \quad \lambda_{2k} = \frac{(-1)^k}{144} (7k+2)((k+1)!)^2. \tag{1.2}$$

so $\lambda_N \neq 0$.

For a polynomial $H = H_N$, let

$$\kappa = H''(1), \quad a_0 = -\frac{1 + \kappa}{2 + \kappa}.$$

The construction below uses $\kappa \neq -2$, verified in Proposition 2.1. In source coordinates set

$$v = xy, \quad S = x^2z, \quad u = 1 + v, \quad \gamma = 1 + a_0v + S, \quad W = u\gamma,$$

and write $p = H'$ and $q(W) = Wp(W) - H(W)$. Define

$$F_N(x, y, z) = (A, B, C), \quad C = x\gamma, \quad B = \frac{1 + p(W)/\gamma}{x}, \quad A = \frac{u + q(W)/\gamma^2}{x^2}. \quad (1.3)$$

Although (1.3) displays quotients, the endpoint identities for H_N make A and B polynomials over $\mathbb{Q}$.

For a target (A, B, C) , put

$$E_{A,B,C}(W) = H_N(W) - BCW + AC^2. \quad (1.4)$$

and define the complete regular locus

$$\Omega_N = \{(A, B, C) \in \mathbb{R}^3 : C \neq 0, \text{ Disc}_W(E_{A,B,C}) \neq 0\}. \quad (1.5)$$

Its connected components will be called real target chambers.

Theorem 1.1 (Exact chamber spectrum). *For every $N \geq 3$, the map F_N in (1.3) is a polynomial Keller map over $\mathbb{Q}$, with*

$$\det DF_N = 1, \quad \deg_{\text{gen}} F_N = N.$$

On Ω_N , the possible real-fiber cardinalities are exactly

$$\boxed{N, N - 2, N - 4, \dots, N \bmod 2}. \quad (1.6)$$

Each cardinality in (1.6) occurs on a nonempty open chamber. Every chamber of Ω_N contains a rational target. Moreover, there are chambers carrying the successive values in (1.6) which form an adjacency chain

$$N \text{ --- } N - 2 \text{ --- } \dots \text{ --- } (N \bmod 2),$$

each edge crossing a single smooth fold of the discriminant of (1.4).

Corollary 1.2 (Migus refinement). *For every even $N \geq 4$, $F_N(\mathbb{R}^3)$ is not Euclidean dense and F_N has generic degree N . Thus the realization half of Migus's classification can be chosen so that the complete regular real-fiber spectrum is $N, N - 2, \dots, 0$, with rational witnesses and a full fold adjacency chain.*

2 The explicit weighted maps

Proposition 2.1. *The polynomials H_N satisfy*

$$H_N(0) = H'_N(0) = H_N(1) = 0, \quad H'_N(1) = -1, \quad H''_N(1) \neq -2.$$

Consequently (1.3) is polynomial, has Jacobian determinant one, and its inverse equation over $C \neq 0$ is (1.4). Every simple root of (1.4) reconstructs one and only one source point.

Proof. The paired factors in (1.1) give $G_N(1) = G'_N(0)$; the first four identities follow immediately from the definition of H_N . Logarithmic differentiation gives

$$H''_{2k+1}(1) = -\frac{2(k+1)}{k} \sum_{i=1}^k \frac{1}{i^2} < -2.$$

In even degree, if

$$D = \frac{5}{12} + \frac{1}{k+2}, \quad Q = \frac{29}{18} + \sum_{j=5}^{k+2} \left(\frac{1}{(j-1)^2} + \frac{1}{j^2} \right),$$

then

$$H''_{2k}(1) = \frac{D^2 - Q}{1 - D} < -2,$$

because $5/12 < D \leq 2/3$ and $Q \geq 29/18 > 1 + (7/12)^2 \geq 1 + (1 - D)^2$.

The divisibilities forced by the endpoint identities cancel the displayed denominators in (1.3). The standard weighted Jacobian calculation gives $\det DF_N = 1$. For completeness, its essential identities are

$$BC = H'_N(W) + \gamma, \quad AC^2 = W(H'_N(W) + \gamma) - H_N(W).$$

Eliminating γ yields (1.4). Conversely, at a simple root W ,

$$\gamma = BC - H'_N(W) = -E'_{A,B,C}(W), \quad x = C/\gamma,$$

and then $y = (W - \gamma)/C$, while z follows uniquely from $\gamma = 1 + a_0xy + x^2z$. If $C \neq 0$, all these quantities are finite exactly when the root is simple. This gives the claimed bijection. $\square$

Taking

$$(A, B, C) = \left(0, -\frac{G'_N(0)}{\lambda_N}, 1 \right)$$

turns (1.4) into $G_N(W)/\lambda_N$, which has N simple roots. Proposition 2.1 therefore also proves that the generic degree is N .

3 A real pencil lemma

The real geometry is isolated in the following elementary statement.

Proposition 3.1 (Pencil spectrum). *Let $H \in \mathbb{R}[W]$ have degree $N \geq 2$, and consider*

$$E_{s,t}(W) = H(W) - sW + t.$$

If one member E_{s_0,t_0} has N distinct real roots, then the root counts on the complement of its discriminant are exactly

$$N, N-2, \dots, N \bmod 2.$$

Arbitrarily close to s_0 there is a vertical line $s = s_1$ which meets chambers with these counts in order, consecutive chambers being separated by one smooth fold. If the input data are rational, s_1 and a target in each selected chamber may be chosen rational.

Proof. Off the discriminant all roots are simple, and nonreal roots occur in conjugate pairs. Hence every root count has the parity of N .

Translate t so that $t_0 = 0$, and put $P_s(W) = H(W) - sW$. Having N simple real zeros is open in s . In a sufficiently small interval about s_0 , avoid the finitely many slopes for which P_s has a degenerate critical point or two critical points with the same critical value. Finiteness follows from the normalization of the discriminant,

$$r \longmapsto (H'(r), rH'(r) - H(r)) :$$

cusps and multiple normalization fibers occur over only finitely many points. Thus a suitable s_1 exists, and it may be taken rational when the data are rational.

Rolle's theorem gives $N - 1$ real critical points of P_{s_1} ; all are nondegenerate and have distinct critical values. Multiply the polynomial by the sign of its leading coefficient, which does not change its roots. Starting from level zero and translating vertically in the direction which passes the local minima, each crossing removes exactly two real roots. The critical values are distinct, so the losses occur one pair at a time. There are $\lfloor N/2 \rfloor$ such crossings. At the end there are no real roots in even degree and one in odd degree. The open intervals between successive critical values give every asserted count. Each interval contains a rational translation.

At a crossed wall the polynomial has one double root r , with $E''(r) \neq 0$, and all other roots are simple. In local coordinates the incidence is

$$\alpha + \beta(W - r)^2 + \text{higher terms}, \quad \alpha\beta \neq 0,$$

which is a smooth fold of the pencil projection. □

4 Proof of the spectrum and rational witnesses

Proof of Theorem 1.1. On a slice $C = C_0 \neq 0$, the target-coordinate change

$$(A, B, C_0) \longmapsto (s, t) = (BC_0, AC_0^2)$$

is a real analytic isomorphism. Proposition 2.1 identifies the real source fiber over every point of Ω_N with the real roots of $H_N(W) - sW + t$.

At the rational target

$$(s_0, t_0) = \left(-\frac{G'_N(0)}{\lambda_N}, 0 \right)$$

this polynomial is G_N/λ_N , with N distinct integer roots. Proposition 3.1 now proves the spectrum, nonempty chambers, rational witnesses for one chamber of every sheet number, and the fold adjacency chain.

The fiber count is locally constant throughout Ω_N , since all roots are simple there. Every connected component of Ω_N is a nonempty open subset of $\mathbb{R}^3$, and $\mathbb{Q}^3$ is dense in $\mathbb{R}^3$; consequently every chamber, not merely the selected chain, contains a rational target. □

Proof of Corollary 1.2. If N is even, Theorem 1.1 supplies a nonempty open chamber on which the real fiber is empty. This chamber is disjoint from $F_N(\mathbb{R}^3)$, so the image is not Euclidean dense. The generic degree was computed in Proposition 2.1. □

Remark 4.1. *The folds in Theorem 1.1 belong to the projection of the one-variable incidence (1.4), not to the Keller map itself. The latter has no critical points. At a wall, the two colliding polynomial roots satisfy $\gamma = -E'(W) = 0$, so their reconstructed source points escape to infinity.*

Credit and provenance

The division of contributions used in this note is as follows. Gallagher developed the all-degree weighted family and exhibited the first quartic empty-real-fiber example [2]; all-degree families also appear in the earlier base manuscript [1]. Migus classified the possible generic degrees of real Keller maps with non-dense Euclidean image [3]. The new result here is the complete parity chamber spectrum, rational target witnesses, and a fold-adjacency chain for one explicit rational map in every degree.

Verification and declaration of generative AI use

The exact arithmetic construction and audited sample witnesses can be checked with

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.venv/bin/python scripts/verify_real_fiber_spectrum.py --show-witnesses
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The author used OpenAI's ChatGPT and Codex during proof organization, adversarial checking, and preparation of this manuscript. The author reviewed the mathematical claims and assumes full responsibility for the contents.

References

- [1] *A counterexample to the Jacobian Conjecture*, anonymous manuscript, 20 July 2026, <https://www.ulam.ai/research/jacobian.pdf>.
- [2] A. Gallagher, *The Jacobian counterexample, explained*, online note, updated 22 July 2026, <https://jacobianfun.org/jacobian-explained>.
- [3] P. Migus, *Generic degrees of real polynomial Keller maps with non-dense image*, arXiv:2607.21572 [math.AG], 2026, <https://arxiv.org/abs/2607.21572>.